This project aims to take data from individual National Football League (NFL) games over the past ten seasons to predict the outcome of future games. Data includes offensive and defensive metrics such as total offensive yards, yards allowed by defense, turnovers, and many more metrics to measure team performance and efficiency.

The overall plan is for each team member to develop a model, write a section of the report on it, and then have a team meeting to discuss individual findings. From there, the team will determine which model appears to have the most overall accuracy and make it the primary focus of the conclusion. Any other notable findings will be mentioned as well, such as if a certain model favors offensive or defensive metrics. The team has a set meeting time of Monday weeknights, with impromptu meetings as needed.

Weekly Timeline

- Week 0 (April 6th)
  - Determine models used
  - Determine who develops a given model
- Week 1 (April 13th)
  - Data collection and cleaning
  - Begin development of models
    - Dylan – Neural Network
    - Dilen – Naïve Bayes
    - Simon – Random Forest
    - Gregory – Logistic Regression
- Week 2 (April 20th)

- Individual report sections detailing findings of given model
- Discuss which model is most overall accurate and notable findings from other models

- Week 3 (April 27th)
  - Polish report and finalize conclusion
- Week 4 (May 4th)
  - Project presentation